

Lab 1: Create Git Repository and Perform Git Operations (Push, Fetch, Merge, Pull, Reset, Revert, Clone, Fork)

Step 1: Install Git

Verify:

```
git --version
```

Step 2: Create GitHub Account

Students create account at:

<https://github.com>

Step 3: Configure Git

```
git config --global user.name "Bhaskar"  
git config --global user.email "bhaskar@gmail.com"
```

Verify:

```
git config --global --list
```

Step 4: Create First Repository

```
mkdir git-demo  
cd git-demo
```

```
git init
```

Create file:

```
echo "# My Project" > README.md
```

Check status:

```
git status
```

Step 5: Git Lifecycle Demonstration

Working Directory

Create:

```
index.html
```

Check:

```
git status
```

Step 6: Staging Area

```
git add .
```

Check:

```
git status
```

Step 7: Local Repository

```
git commit -m "Initial Commit"
```

Check:

```
git log
```

Step 8: GitHub Repository

Create repository in GitHub:

```
git-demo
```

Connect:

```
git remote add origin https://github.com/bhaskarmarapelli/gitdemo.git
```

Push:

```
git push -u origin main
```

Step 9: Make a remote change and fetch it

Edit index.html directly on GitHub.

Example:

```
<h2>Updated from GitHub</h2>
```

In local terminal:

```
git fetch origin
```

Check fetched commits:

```
git log --oneline --all
```

Step 10: Merge fetched changes

```
git merge origin/main
```

Now the local repository contains the remote changes.

Step 11: Pull latest changes

Make another change on GitHub.

Run:

```
git pull origin main
```

This performs fetch + merge automatically.

Step 12: Reset a local commit

Modify index.html.

```
<p>Temporary change</p>
```

Commit:

```
git add .  
git commit -m "Temporary commit"
```

Undo the commit but keep changes:

```
git reset --soft HEAD~1
```

Undo commit and discard changes:

```
git reset --hard HEAD~1
```

Step 13: Revert a committed change

Create a new commit.

```
git add .  
git commit -m "Add paragraph"
```

Undo it while preserving history:

```
git revert HEAD
```

Git creates a new commit that reverses the previous commit.

Step 14: Clone the repository

```
cd ..
```

```
git clone https://github.com/bhaskarmarapelli/gitdemo.git
```

A new folder gitdemo is created.

Step 15: Fork a public repository

On GitHub open a public repository.

Click Fork.

Clone the fork:

```
git clone https://github.com/your-username/forked-repository.git
```